

# PROFORMA — High-Traffic Client Project Infrastructure Billing

LIONGATE SARL — Infrastructure Cost Estimation

Document: PROFORMA-03

Context: High-Traffic Client Project (Public SaaS / E-commerce / Multi-tenant Platform)

Hosting Provider: Hetzner Cloud (+ Cloudflare edge)

Currency: EUR (€)

Version: 1.0

## How to read this document:

This proforma covers infrastructure costs for client projects that must serve **high concurrent traffic**, with strong **availability, performance, and security guarantees** (SLA-backed).

Four tiers are presented, scaling from "launching public product" to "regional-scale platform".

## Reference Stack

| Layer            | Technology                    | Notes                                  |
|------------------|-------------------------------|----------------------------------------|
| Edge / CDN / WAF | Cloudflare Pro/Business       | DDoS, caching, WAF, image optimization |
| Load Balancer    | Hetzner LB21 / LB31           | TCP+HTTP, multi-target, health checks  |
| App tier         | Node.js / NestJS / Next.js    | Horizontally scaled (≥2 nodes)         |
| DB primary       | PostgreSQL 16 (dedicated CPU) | CCX-series, tuned for OLTP             |
| DB replica       | PostgreSQL streaming replica  | Read scaling + standby                 |
| Cache / Queue    | Redis 7                       | Sessions, queue, rate limiting         |

| Layer          | Technology                                 | Notes                               |
|----------------|--------------------------------------------|-------------------------------------|
| Object storage | Hetzner Object Storage (S3-compatible)     | Media, exports, uploads             |
| Search (opt.)  | Meilisearch / OpenSearch                   | Product/document search             |
| Observability  | Grafana + Prometheus + Loki + Alertmanager | Dedicated monitoring node           |
| CI/CD          | GitHub Actions + self-hosted runner        | Build + deploy + DB migration gates |
| Secrets        | Doppler / HashiCorp Vault (cloud)          | Production-grade secret management  |

## Tier 1 — Launch Setup (Single-region, scalable)

Production-ready launch for a public client product. Single AZ but split tiers.

**Traffic assumption:** 50k–150k requests/day, 1,000–2,000 concurrent users, < 50 req/s sustained.

### Environment: Production (Launch)

| # | Resource                  | Spec                                | Unit Price | Qty | Monthly (€) | Annual (€) |
|---|---------------------------|-------------------------------------|------------|-----|-------------|------------|
| 1 | App Servers (behind LB)   | CX42 — 8 vCPU, 16 GB RAM            | €17.77     | 2   | 35.54       | 426.48     |
| 2 | DB Server (dedicated CPU) | CCX23 — 4 vCPU dedicated, 16 GB RAM | €22.75     | 1   | 22.75       | 273.00     |
| 3 | Redis Server              | CX32 — 4 vCPU, 8 GB RAM             | €8.21      | 1   | 8.21        | 98.52      |
| 4 | Load Balancer             | LB21 — up to 25 targets             | €10.79     | 1   | 10.79       | 129.48     |

| #  | Resource                   | Spec                           | Unit Price | Qty | Monthly (€) | Annual (€) |
|----|----------------------------|--------------------------------|------------|-----|-------------|------------|
| 5  | Volume — DB data           | 250 GB                         | €13.00     | 1   | 13.00       | 156.00     |
| 6  | Volume — shared uploads    | 200 GB                         | €10.40     | 1   | 10.40       | 124.80     |
| 7  | Private Network (VLAN)     | App ↔ DB ↔ Redis               | Free       | 1   | 0.00        | 0.00       |
| 8  | Storage Box (Backups)      | BX31 — 1 TB                    | €8.69      | 1   | 8.69        | 104.28     |
| 9  | Object Storage (media)     | S3 — 100 GB avg                | €1.20      | 1   | 1.20        | 14.40      |
| 10 | Floating IPs               | Static IPv4 (LB + admin)       | €2.00      | 2   | 4.00        | 48.00      |
| 11 | Monitoring node            | CX22 — Grafana/Prometheus/Loki | €4.35      | 1   | 4.35        | 52.20      |
| 12 | Cloudflare Pro (CDN + WAF) | Per zone                       | €20.00     | 1   | 20.00       | 240.00     |
| 13 | SSL                        | Let's Encrypt wildcard         | Free       | 1   | 0.00        | 0.00       |
| 14 | Domain (amortised)         | Annual                         | €15.00/yr  | 1   | 1.25        | 15.00      |
| 15 | Transactional email        | Brevo Business — 100k/mo       | €49.00     | 1   | 49.00       | 588.00     |
| 16 | External uptime monitoring | UptimeRobot Pro                | €7.00      | 1   | 7.00        | 84.00      |
| 17 | Secrets management         | Doppler Team — 5 seats         | €18.00     | 1   | 18.00       | 216.00     |

| #  | Resource                         | Spec       | Unit Price | Qty | Monthly (€)    | Annual (€)       |
|----|----------------------------------|------------|------------|-----|----------------|------------------|
| 18 | VM<br>Snapshots<br>(rotation ×4) | ~80 GB avg | €0.95      | 1   | 0.95           | 11.40            |
| —  | <b>SUBTOTAL</b>                  |            |            |     | <b>€215.13</b> | <b>€2,581.56</b> |

## Environment: Staging

| # | Resource        | Spec        | Monthly (€)   | Annual (€)     |
|---|-----------------|-------------|---------------|----------------|
| 1 | App+DB host     | CX42        | 17.77         | 213.24         |
| 2 | Redis           | CX22        | 4.35          | 52.20          |
| 3 | Volume          | 100 GB      | 5.20          | 62.40          |
| 4 | Floating IP     | Static IPv4 | 2.00          | 24.00          |
| 5 | Brevo (test)    | Brevo Free  | 0.00          | 0.00           |
| — | <b>SUBTOTAL</b> |             | <b>€29.32</b> | <b>€351.84</b> |

## Environment: Development (Shared)

| # | Resource                | Spec                  | Monthly (€)  | Annual (€)    |
|---|-------------------------|-----------------------|--------------|---------------|
| 1 | Local Docker Compose    | Developer workstation | 0.00         | 0.00          |
| 2 | Shared dev VPS (review) | CX32                  | 8.21         | 98.52         |
| — | <b>SUBTOTAL</b>         |                       | <b>€8.21</b> | <b>€98.52</b> |

## Tier 1 — Total Summary

| Environment | Monthly MIN (€)       | Monthly MAX (€) | Annual MIN (€) | Annual MAX (€) |
|-------------|-----------------------|-----------------|----------------|----------------|
| Production  | 215.13                | 247.40          | 2,581.56       | 2,968.79       |
| Staging     | 11.73 <i>(paused)</i> | 29.32           | 140.76         | 351.84         |

| Environment         | Monthly MIN (€) | Monthly MAX (€) | Annual MIN (€)   | Annual MAX (€)   |
|---------------------|-----------------|-----------------|------------------|------------------|
| Development         | 0.00            | 8.21            | 0.00             | 98.52            |
| <b>TOTAL TIER 1</b> | <b>€226.86</b>  | <b>€284.93</b>  | <b>€2,722.32</b> | <b>€3,419.15</b> |

## Tier 2 – Growth Setup (Replica + autoscale-ready)

Read replica added, dedicated app autoscaling pool, hardened observability.

**Traffic assumption:** 150k–500k requests/day, 3,000–6,000 concurrent users, ~150 req/s sustained.

### Environment: Production (Growth)

| # | Resource                   | Spec                                | Unit Price | Qty | Monthly (€) | Annual (€) |
|---|----------------------------|-------------------------------------|------------|-----|-------------|------------|
| 1 | App Servers (behind LB)    | CX42 — 8 vCPU, 16 GB RAM            | €17.77     | 3   | 53.31       | 639.72     |
| 2 | DB Primary (dedicated CPU) | CCX33 — 8 vCPU dedicated, 32 GB RAM | €45.50     | 1   | 45.50       | 546.00     |
| 3 | DB Read Replica            | CCX23 — 4 vCPU dedicated, 16 GB RAM | €22.75     | 1   | 22.75       | 273.00     |
| 4 | Redis (primary + replica)  | CX32                                | €8.21      | 2   | 16.42       | 197.04     |
| 5 | Load Balancer              | LB31 — up to 75 targets             | €18.04     | 1   | 18.04       | 216.48     |
| 6 | Volume — DB data (primary) | 500 GB                              | €26.00     | 1   | 26.00       | 312.00     |
| 7 | Volume — DB data (replica) | 500 GB                              | €26.00     | 1   | 26.00       | 312.00     |

| #  | Resource                         | Spec                     | Unit Price | Qty | Monthly (€)    | Annual (€)       |
|----|----------------------------------|--------------------------|------------|-----|----------------|------------------|
| 8  | Volume — shared uploads          | 500 GB                   | €26.00     | 1   | 26.00          | 312.00           |
| 9  | Object Storage (media + exports) | S3 — 500 GB avg          | €6.00      | 1   | 6.00           | 72.00            |
| 10 | Storage Box (Backups)            | BX41 — 5 TB              | €15.21     | 1   | 15.21          | 182.52           |
| 11 | Floating IPs                     | Static IPv4              | €2.00      | 2   | 4.00           | 48.00            |
| 12 | Monitoring node                  | CX32 — Grafana stack     | €8.21      | 1   | 8.21           | 98.52            |
| 13 | Log retention (Loki) volume      | 200 GB                   | €10.40     | 1   | 10.40          | 124.80           |
| 14 | Self-hosted CI runner            | CX42                     | €17.77     | 1   | 17.77          | 213.24           |
| 15 | Cloudflare Business              | Per zone                 | €185.00    | 1   | 185.00         | 2,220.00         |
| 16 | Transactional email              | Brevo Business — 100k/mo | €49.00     | 1   | 49.00          | 588.00           |
| 17 | External uptime + APM            | Better Stack / Checkly   | €25.00     | 1   | 25.00          | 300.00           |
| 18 | Secrets management               | Doppler Team — 10 seats  | €36.00     | 1   | 36.00          | 432.00           |
| 19 | Domain (amortised, 2 zones)      | Annual                   | €30.00/yr  | 1   | 2.50           | 30.00            |
| 20 | VM Snapshots                     | ~150 GB avg              | €1.80      | 1   | 1.80           | 21.60            |
| —  | <b>SUBTOTAL</b>                  |                          |            |     | <b>€594.91</b> | <b>€7,138.92</b> |

## Environment: Staging (close-to-prod)

| # | Resource    | Spec        | Monthly (€) | Annual (€) |
|---|-------------|-------------|-------------|------------|
| 1 | App tier    | CX42 ×1     | 17.77       | 213.24     |
| 2 | DB          | CX42        | 17.77       | 213.24     |
| 3 | Redis       | CX22        | 4.35        | 52.20      |
| 4 | Volume      | 200 GB      | 10.40       | 124.80     |
| 5 | Floating IP | Static IPv4 | 2.00        | 24.00      |
| — | SUBTOTAL    |             | €52.29      | €627.48    |

## Tier 2 — Total Summary

| Environment  | Monthly MIN (€)       | Monthly MAX (€) | Annual MIN (€) | Annual MAX (€) |
|--------------|-----------------------|-----------------|----------------|----------------|
| Production   | 594.91                | 684.15          | 7,138.92       | 8,209.76       |
| Staging      | 20.92 <i>(paused)</i> | 52.29           | 251.04         | 627.48         |
| Development  | 0.00                  | 8.21            | 0.00           | 98.52          |
| TOTAL TIER 2 | €615.83               | €744.65         | €7,389.96      | €8,935.76      |

## Tier 3 — Scale Setup (HA, multi-AZ-style)

Active-passive DB failover, multiple Redis nodes, dedicated runners, full observability, blue/green deploys, hardened DR.

**Traffic assumption:** 500k–2M requests/day, 10k+ concurrent users, ~500 req/s sustained.

## Environment: Production (Scale)

| #  | Resource                       | Spec                             | Unit Price | Qty | Monthly (€) | Annual (€) |
|----|--------------------------------|----------------------------------|------------|-----|-------------|------------|
| 1  | App Servers (autoscaling pool) | CX52 — 16 vCPU, 32 GB RAM        | €35.34     | 4   | 141.36      | 1,696.32   |
| 2  | DB Primary (dedicated CPU)     | CCX43 — 16 vCPU dedicated, 64 GB | €91.00     | 1   | 91.00       | 1,092.00   |
| 3  | DB Standby + Replica           | CCX33 — 8 vCPU dedicated, 32 GB  | €45.50     | 2   | 91.00       | 1,092.00   |
| 4  | Redis Cluster (3 nodes)        | CX32                             | €8.21      | 3   | 24.63       | 295.56     |
| 5  | Load Balancer                  | LB31 — up to 75 targets          | €18.04     | 2   | 36.08       | 432.96     |
| 6  | Volume — DB data               | 1 TB                             | €52.00     | 3   | 156.00      | 1,872.00   |
| 7  | Volume — shared uploads        | 1 TB                             | €52.00     | 1   | 52.00       | 624.00     |
| 8  | Object Storage                 | S3 — 2 TB avg                    | €24.00     | 1   | 24.00       | 288.00     |
| 9  | Storage Box (Backups)          | BX41 — 5 TB                      | €15.21     | 2   | 30.42       | 365.04     |
| 10 | Floating IPs                   | Static IPv4                      | €2.00      | 3   | 6.00        | 72.00      |
| 11 | Monitoring nodes (HA)          | CX32                             | €8.21      | 2   | 16.42       | 197.04     |
| 12 | Log retention (Loki) volume    | 500 GB                           | €26.00     | 1   | 26.00       | 312.00     |
| 13 | Self-hosted CI runners         | CX42                             | €17.77     | 2   | 35.54       | 426.48     |



| #  | Resource                       | Spec                       | Unit Price | Qty | Monthly (€)      | Annual (€)        |
|----|--------------------------------|----------------------------|------------|-----|------------------|-------------------|
| 14 | Cloudflare Business + Argo     | Per zone + smart routing   | €235.00    | 1   | 235.00           | 2,820.00          |
| 15 | Transactional email            | Brevo Enterprise — 1M/mo   | €179.00    | 1   | 179.00           | 2,148.00          |
| 16 | External monitoring + APM      | Better Stack + Sentry Team | €75.00     | 1   | 75.00            | 900.00            |
| 17 | Secrets management             | Doppler Pro — 15 seats     | €120.00    | 1   | 120.00           | 1,440.00          |
| 18 | Domain (multi-zone, amortised) | Annual                     | €60.00/yr  | 1   | 5.00             | 60.00             |
| 19 | VM Snapshots                   | ~300 GB avg                | €3.60      | 1   | 3.60             | 43.20             |
| —  | <b>SUBTOTAL</b>                |                            |            |     | <b>€1,348.05</b> | <b>€16,176.60</b> |

### Tier 3 — Total Summary

| Environment         | Monthly MIN (€)       | Monthly MAX (€)  | Annual MIN (€)    | Annual MAX (€)    |
|---------------------|-----------------------|------------------|-------------------|-------------------|
| Production          | 1,348.05              | 1,550.26         | 16,176.60         | 18,603.09         |
| Staging             | 30.00 <i>(paused)</i> | 80.00            | 360.00            | 960.00            |
| Development         | 0.00                  | 8.21             | 0.00              | 98.52             |
| <b>TOTAL TIER 3</b> | <b>€1,378.05</b>      | <b>€1,638.47</b> | <b>€16,536.60</b> | <b>€19,661.61</b> |

### Tier 4 — Enterprise / Regional Scale

Multi-region active/active or large monolithic dedicated cluster, dedicated database servers, contractual SLAs, dedicated SRE tooling.

**Traffic assumption:** 2M+ requests/day, 25k+ concurrent users, > 1,000 req/s sustained.

## Environment: Production (Enterprise)

| #  | Resource                           | Spec                                     | Unit Price | Qty | Monthly (€) | Annual (€) |
|----|------------------------------------|------------------------------------------|------------|-----|-------------|------------|
| 1  | Dedicated server (Robot AX102)     | Ryzen 9, 128 GB, 2× NVMe — app pool host | €119.00    | 3   | 357.00      | 4,284.00   |
| 2  | Dedicated server (Robot EX144)     | DB host — Xeon, 256 GB RAM, NVMe RAID    | €235.00    | 2   | 470.00      | 5,640.00   |
| 3  | Redis Cluster (HA, 6 nodes)        | CX42                                     | €17.77     | 6   | 106.62      | 1,279.44   |
| 4  | Load Balancers (multi-zone)        | LB31                                     | €18.04     | 4   | 72.16       | 865.92     |
| 5  | Object Storage                     | S3 — 10 TB avg                           | €120.00    | 1   | 120.00      | 1,440.00   |
| 6  | Storage Box (DR backups)           | BX51 — 10 TB                             | €30.43     | 2   | 60.86       | 730.32     |
| 7  | Floating IPs                       | Static IPv4                              | €2.00      | 6   | 12.00       | 144.00     |
| 8  | Dedicated monitoring + APM stack   | CX42 ×3 + storage                        | €17.77     | 3   | 53.31       | 639.72     |
| 9  | Self-hosted CI farm                | CX52                                     | €35.34     | 3   | 106.02      | 1,272.24   |
| 10 | Cloudflare Enterprise (negotiated) | Per zone — contract                      | €1,500.00  | 1   | 1,500.00    | 18,000.00  |
| 11 | Transactional email                | SendGrid / Postmark — 5M/mo              | €450.00    | 1   | 450.00      | 5,400.00   |

| #  | Resource                               | Spec       | Unit Price | Qty | Monthly (€)      | Annual (€)        |
|----|----------------------------------------|------------|------------|-----|------------------|-------------------|
| 12 | Observability SaaS (Datadog/New Relic) | Per host   | €600.00    | 1   | 600.00           | 7,200.00          |
| 13 | Secrets / IAM (Vault Cloud)            | Tier: Plus | €350.00    | 1   | 350.00           | 4,200.00          |
| 14 | Domain (multi-region, amortised)       | Annual     | €150.00/yr | 1   | 12.50            | 150.00            |
| 15 | On-call paging (PagerDuty)             | 10 seats   | €210.00    | 1   | 210.00           | 2,520.00          |
| —  | <b>SUBTOTAL</b>                        |            |            |     | <b>€4,480.47</b> | <b>€53,765.64</b> |

## Tier 4 — Total Summary

| Environment         | Monthly MIN (€)  | Monthly MAX (€)  | Annual MIN (€)    | Annual MAX (€)    |
|---------------------|------------------|------------------|-------------------|-------------------|
| Production          | 4,480.47         | 5,152.54         | 53,765.64         | 61,830.49         |
| Staging (large)     | 150.00           | 250.00           | 1,800.00          | 3,000.00          |
| Development         | 8.21             | 16.42            | 98.52             | 197.04            |
| <b>TOTAL TIER 4</b> | <b>€4,638.68</b> | <b>€5,418.96</b> | <b>€55,664.16</b> | <b>€65,027.53</b> |

## High-Traffic — Tier Comparison Dashboard

| Tier      | Setup Type                   | Monthly MIN | Monthly MAX | Annual MIN | Annual MAX | Best for              |
|-----------|------------------------------|-------------|-------------|------------|------------|-----------------------|
| <b>T1</b> | Launch — single region split | €226.86     | €284.93     | €2,722     | €3,419     | Public product launch |

| Tier | Setup Type                        | Monthly MIN | Monthly MAX | Annual MIN | Annual MAX | Best for                      |
|------|-----------------------------------|-------------|-------------|------------|------------|-------------------------------|
| T2   | Growth — read replica + autoscale | €615.83     | €744.65     | €7,390     | €8,936     | Active product, 100k+ users   |
| T3   | Scale — HA + DR                   | €1,378.05   | €1,638.47   | €16,537    | €19,662    | Mission-critical SaaS         |
| T4   | Enterprise — dedicated + multi-LB | €4,638.68   | €5,418.96   | €55,664    | €65,028    | Regional platform, SLA-backed |

## Hidden & Variable Costs

| Cost Item                             | Frequency | Estimated Amount      | Notes                                |
|---------------------------------------|-----------|-----------------------|--------------------------------------|
| Cloudflare egress / Workers           | Monthly   | €0–€200               | Depends on traffic shape             |
| Egress to internet (Hetzner)          | Monthly   | €0–€50                | Includes 20 TB free per cloud server |
| DB IOPS / volume burst                | Monthly   | +€10–€80              | If volume sized too small            |
| Penetration test (annual)             | Annual    | €1,500–€6,000         | Required by client SLA               |
| Compliance audit (SOC2 / ISO)         | Annual    | €5,000–€25,000        | Only if mandated                     |
| 3rd-party APIs (Stripe, Twilio, etc.) | Monthly   | Variable              | Charged to client                    |
| Sentry / error tracking overage       | Monthly   | +€26–€80              | Past free quota                      |
| Backup restoration test (quarterly)   | Quarterly | €0 (engineering time) | Recommended for DR validation        |

| Cost Item                       | Frequency | Estimated Amount | Notes                                |
|---------------------------------|-----------|------------------|--------------------------------------|
| Domain SSL EV / commercial cert | Annual    | €100–€500        | Only if mandated                     |
| 24/7 on-call premium            | Monthly   | +€500–€2,500     | If client SLA requires 24/7 response |

# Budget Planning Summary (High-Traffic Client Project)

LIONGATE SARL – High-Traffic Client Project Budget Reference

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LAUNCH MONTHLY BUDGET:           €226.86    (Tier 1 – single region)  
GROWTH MONTHLY BUDGET:       €615.83    (Tier 2 – replica + autoscale)  
SCALE MONTHLY BUDGET:        €1,378.05 (Tier 3 – HA + DR)  
ENTERPRISE MONTHLY BUDGET:   €4,638.68 (Tier 4 – dedicated/regional)

LAUNCH ANNUAL BUDGET:        €2,722.32  
GROWTH ANNUAL BUDGET:        €7,389.96  
SCALE ANNUAL BUDGET:         €16,536.60  
ENTERPRISE ANNUAL BUDGET:    €55,664.16

CONTINGENCY BUFFER (+20% for high-traffic uncertainty):  
Tier 2 growth:       €7,389.96 × 1.20 = €8,867.95/yr  
Tier 3 scale:        €16,536.60 × 1.20 = €19,843.92/yr  
Tier 4 enterprise:€55,664.16 × 1.20 = €66,797.00/yr

# Decision Guide – Which tier to quote a client?

| Question                                      | If YES → recommend |
|-----------------------------------------------|--------------------|
| Public launch, < 2k concurrent users expected | Tier 1             |
| Active SaaS, recurring growth, read-heavy     | Tier 2             |

| Question                                          | If YES → recommend |
|---------------------------------------------------|--------------------|
| 24/7 critical, contractual uptime $\geq$ 99.9%    | Tier 3             |
| Multi-region, regulated industry, SLA-backed      | Tier 4             |
| Client requires dedicated CPU + dedicated backups | Tier 2 minimum     |
| Client requires WAF + DDoS guarantees             | Tier 1 minimum     |

*Document: PROFORMA-03 | Context: High-Traffic Client Project*

*Prices: Hetzner Cloud + Hetzner Robot published rates + Cloudflare/Brevo/Doppler list prices*

*Review: Quarterly, or per client contract signing*

*Prepared by: LIONGATE SARL Engineering Team*